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Subject: Business Analytics

MBA-Regular (2024-2026)

Sem: II Section: II

Assignment-I

Roll-0026

**Q. From the uploaded Excel dataset houseprices (File name), identify few valid objectives from Promoter and Buyers point of view. Use graphical representations to fulfill objectives.**

Ans. The provided Excel dataset have total 128 no. of data. The dataset has several attributes like – Price, SqFt, Neighborhood, Bedrooms, Bathrooms, Offers, Bricks etc.

Now the objectives to extract information from the given dataset are:

A. Promoter Point of View:

- a. Which Neighborhood have higher and lower price range.
- b. Which Neighborhood should be given more offers.
- c. Most valued 5 properties. Least valued 5 properties.
- d. Relationship between Price and Offer.
- e. Co-Relationship between Price and Neighborhood.
- f. Price/SqFt vs Price/SqFt, Neighborhood wise.
- g. Prediction of Price on the future projects, Neighborhood wise.

B. Buyers Point of View:

- a. Price/SqFt, Neighborhood wise.
- b. Average Price Neighborhood wise.
- c. Median Price Neighborhood wise.
- d. Neighborhood Vs Offers Relationship.
- e. Offers Vs SqFt relationship.
- f. Comparison of Bedrooms, Neighborhood wise.
- g. Bedroom-Price Relationship

Now, for the analysis of the dataset we have to study the relationships of the following attributes.

### **1. No. of houses based on neighborhood:**

From the above datasheet we have 128 no. of data, of different Neighborhood.

| Neighborhood | No. of House |
|--------------|--------------|
| East         | 45           |
| West         | 44           |
| North        | 39           |

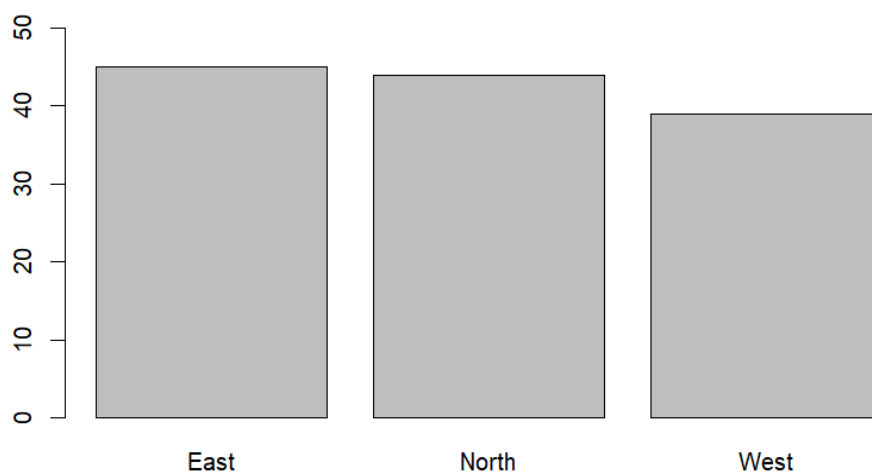
### **R code:**

```
Neighborhood=as.factor(Neighborhood)
```

```
summary(Neighborhood)
```

```
t=table(Neighborhood)
```

```
barplot(t,ylim=c(0,50))
```



From the above observations we can say that the houses are mostly equally distributed among the neighborhoods.

## **2. Average Price of the houses:**

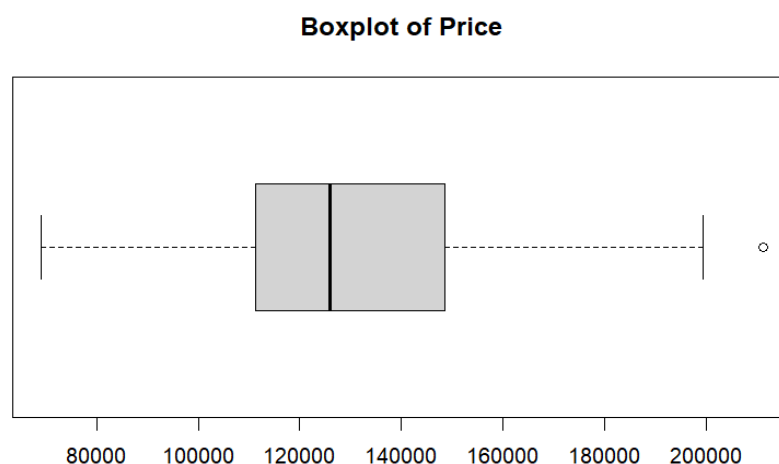
Mean value of the price of the houses in the given dataset is: 130427.30

R code: `mean(Price)`

## **3. Median Price of the houses:**

Median value of the houses in the given dataset is: 125950

R code: `median(Price)`



R code: `boxplot(Price, horizontal = T, main = "Boxplot of Price")`

**Analysis:** The median range of the price is around 125000 and the first two Quartiles of the Price range is less than the last two Quartiles of the price range. Also, there is an outlier of range 200000+ of price.

## **4. Average Price of the houses based on Neighborhood:**

Avg. Price of the houses in the East Neighborhood - 125231.10

R code: `mean(Price[Neighborhood=="East"])`

Avg. Price of the houses in the West Neighborhood - 159294.40

R code: `mean(Price[Neighborhood=="West"])`

Avg. Price of the houses in the North Neighborhood - 110154.5

R code: `mean(Price[Neighborhood=="North"])`

## **5. Median Price of the houses based on Neighborhood:**

Median Price of the houses in the East Neighborhood – 123100

Median Price of the houses in the West Neighborhood - 155400

Median Price of the houses in the North Neighborhood - 110750

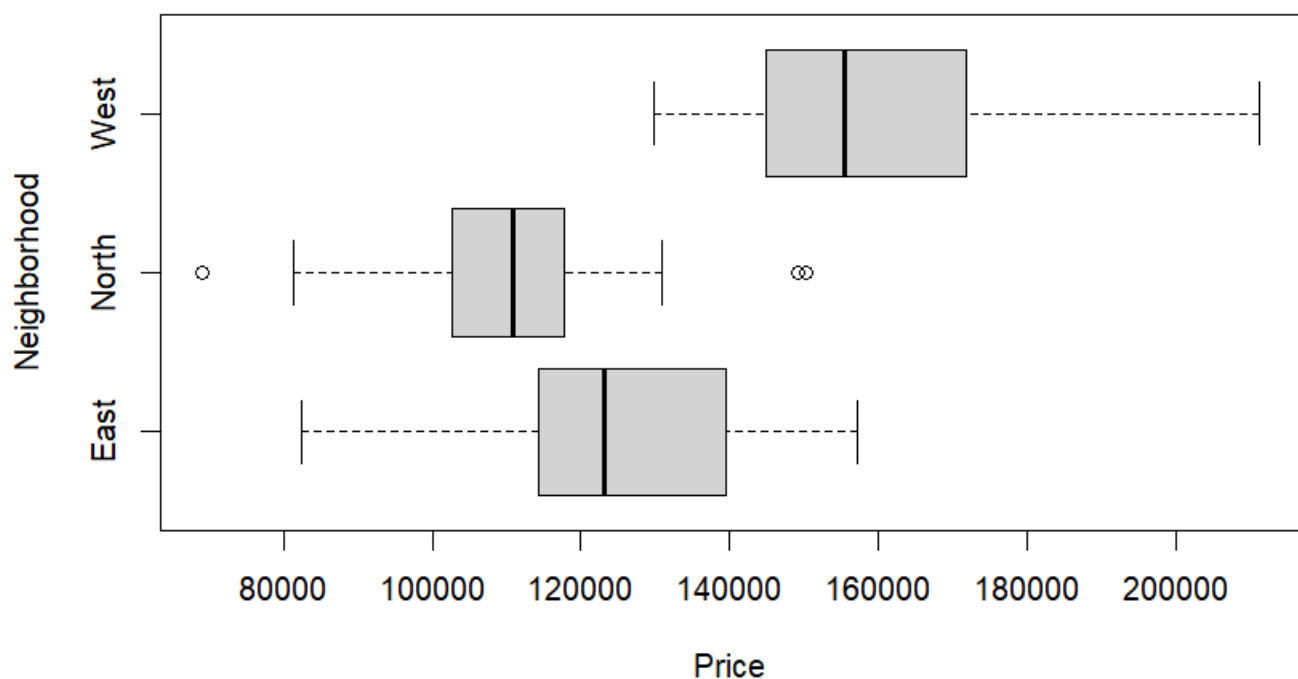
R code: `median(Price[Neighborhood=="East"])`

`median(Price[Neighborhood=="West"])`

`median(Price[Neighborhood=="North"])`

`boxplot(Price~Neighborhood,horizontal = T)`

**Box Plot of Neighborhood Vs Price**



Analysis: From the above observations we can see that West Neighborhood have more price range than the overall price range. The 4<sup>th</sup> Quartile range of West neighborhood is higher than the others. This can be for the reason that west neighborhood houses are situated in the Prior areas of the city and have more demands than the other neighborhood areas. And there exists few outliers of the north neighborhood, suggest that the price range variance of north neighborhood is more.

## **6. Average Price of the houses based on Bricks:**

Average Price of houses having bricks: 147769

R code: `mean(Price[Brick=="Yes"])`

Average Price of houses having no bricks: 121958.1

R code: `mean(Price[Brick=="No"])`

## **7. Median Price of the houses based on Bricks:**

Median price of houses having bricks: 147500

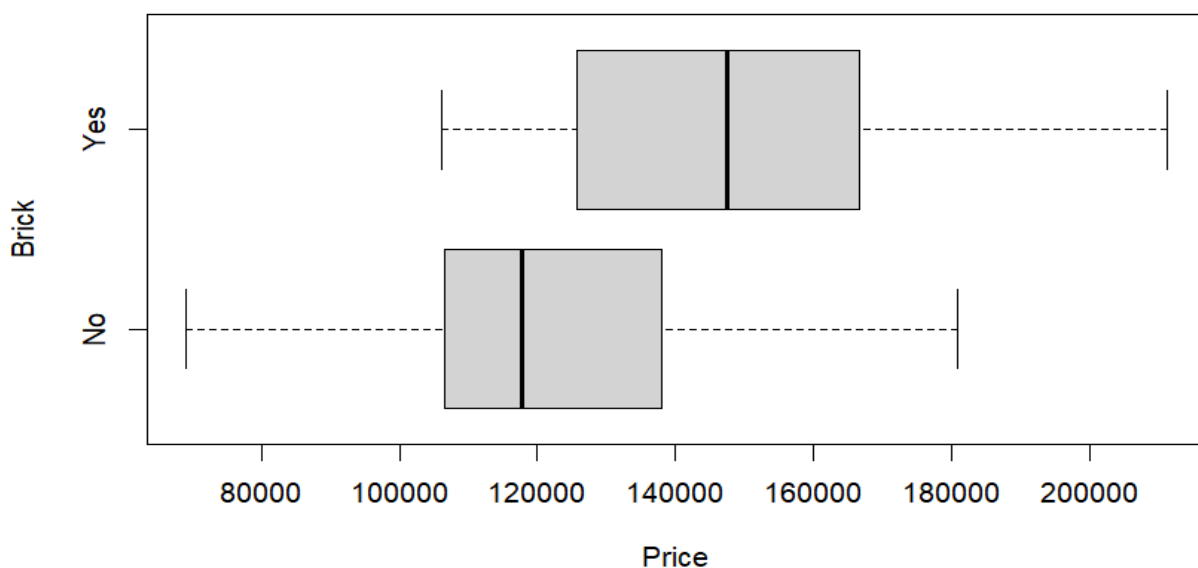
Median price of houses having no bricks: 117650

R code: `median(Price[Brick=="Yes"])`

`median(Price[Brick=="No"])`

`boxplot(Price~Brick, horizontal=T)`

**Box plot of Price vs Brick**



**Analysis:** Price based on bricks are much higher than the price based on non-bricks. For houses based on bricks, the 4<sup>th</sup> Quartile range is higher than the others. This implies that, the brick houses have higher price range. This may be because of brick based houses are located in popular areas and non-bricks houses are located in hill areas which is not so popular area.

## **8. Co-relationship between SqFt and Price:**

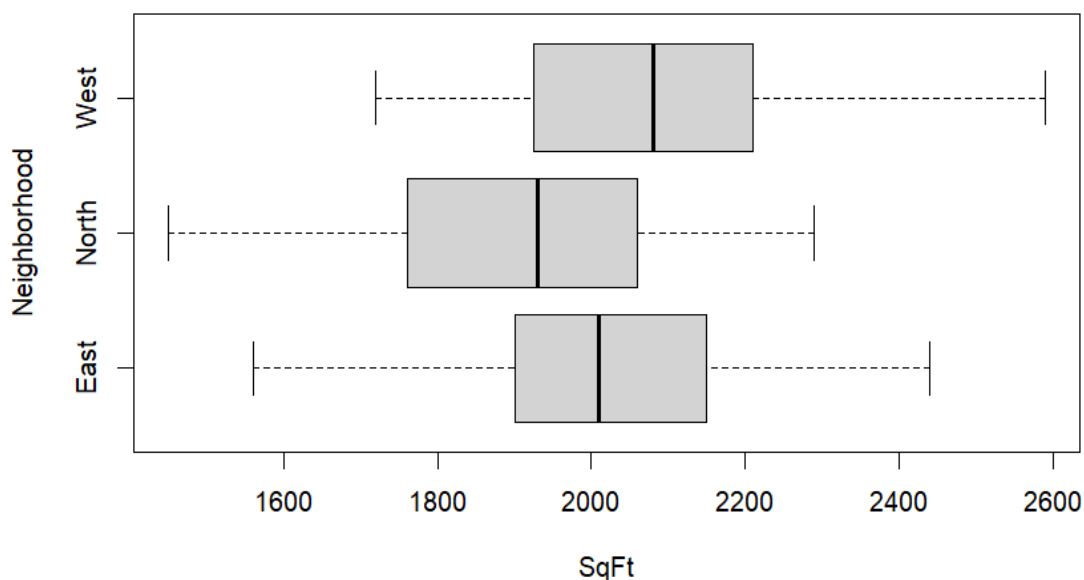
The value of the co-relationship between SqFt and Price is 0.553.

R Code: `cor(SqFt,Price)`

**Analysis:** The value of the co-relationship is positive, and near to 0.5 suggest that the SqFt and Price have positive co-relationship between them. As the SqFt area increases, the price of the houses also increases.

**9. SqFt Vs Neighborhood Relationship:** We can plot the boxplot between SqFt and Neighborhood and can check the relationship between them.

R code: `boxplot(SqFt~Neighborhood,horizontal = T)`



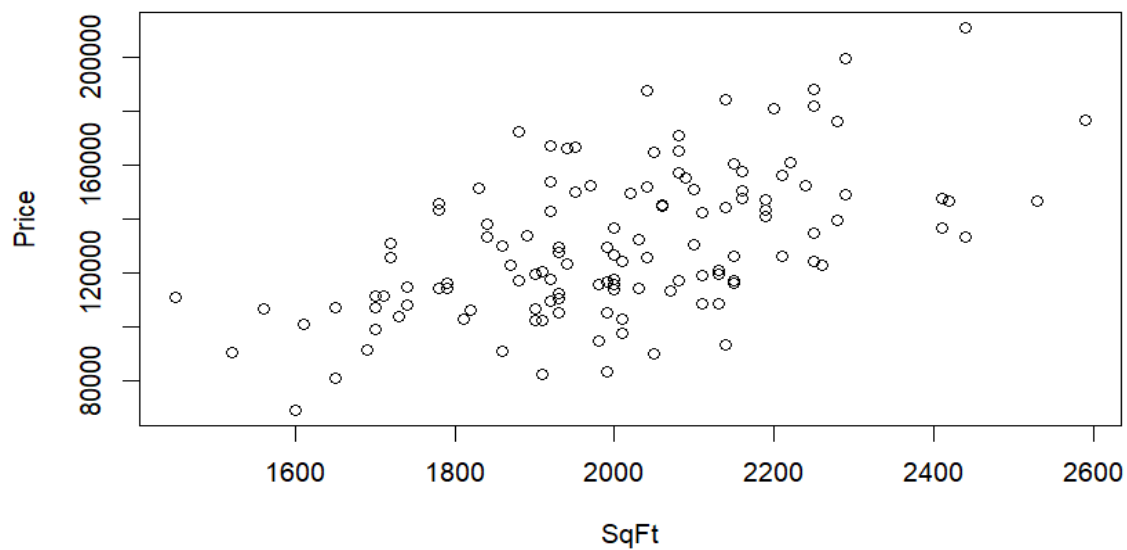
**Analysis:** From the above plot we can say that the west neighborhood has more area sq.ft houses than the other two neighborhoods and the north neighborhood has less area sq.ft houses.

## **10. Price Vs. SqFt relationship:**

As the SqFt value increases, price also get increased. But after a certain range of SqFt, (i.e., 2200 approx.) the variance of the price is high. This may be for the factors of neighborhood, bathrooms, urban or rural areas, underdeveloped and metro cities etc.

R code: `plot(SqFt,Price)`

### Scatterplot between Price and SqFt



#### **11. Most and Least 5 House prices and their neighborhood:**

##### **5 highest House Prices:**

| House Prices | Neighborhood |
|--------------|--------------|
| 211200       | West         |
| 199500       | West         |
| 188300       | West         |
| 188000       | West         |
| 184300       | West         |

R code: `tail(sort(Price),5)`

`which(Price=="211200")`

##### **5 lowest House Prices:**

| House Prices | Neighborhood |
|--------------|--------------|
| 69100        | North        |
| 81300        | North        |
| 82300        | East         |
| 83600        | North        |
| 90300        | North        |

R code: `head(sort(Price),5)`

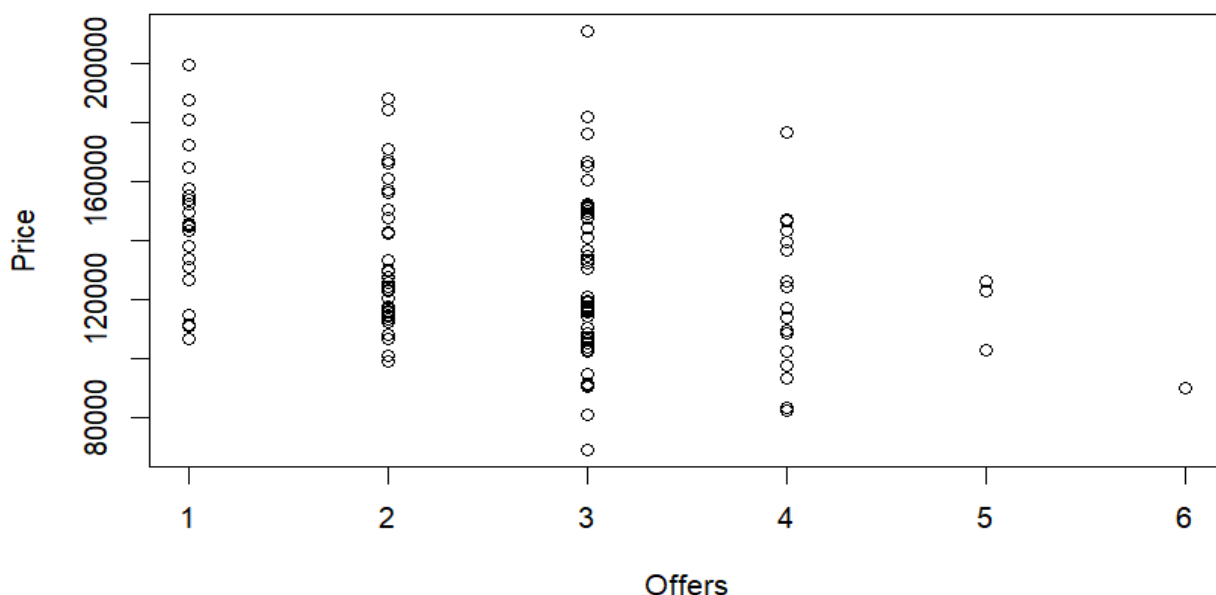
`which(Price=="90300")`

**Analysis:** From the above data we can observe that most of the top house prices are from West neighborhood, and the least house prices are mostly from north neighborhood. This can be because of the reason that west neighborhood has most of the public demand and north neighborhood have least demand may be because of the geographical position.

## **12. Price Vs Offer Relationship:**

As the Price is lower, the offers are more, but where the prices are higher, the offers are less. The co relationship value between Offers and Price is (-0.3136).

R code: `plot(Offers,Price)`  
`cor(Offers,Price)`



**Analysis:** As we can check, the co-relation value between Offers and Price is negative, that implies as the value of the prices increases the number of offers decreases. This is because that, the houses which are in prior location have more demands and easily sold out with less number of offers where as the houses that located in the rural areas have less demand and to sell them the promoter have to provide more numbers of offers with less price.

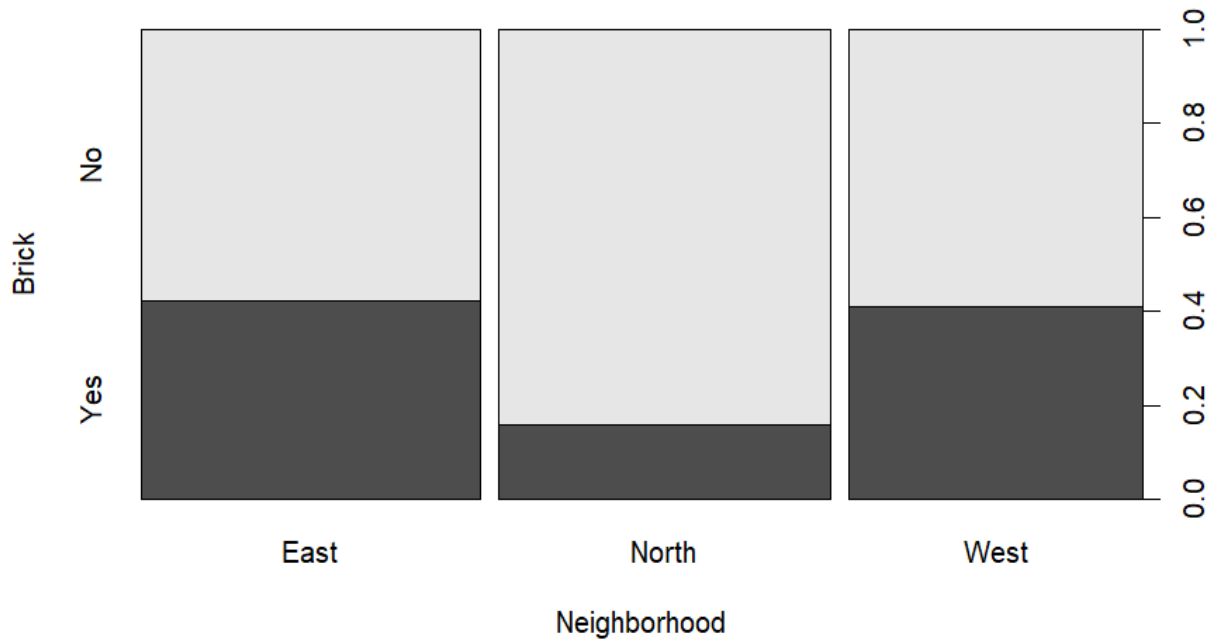


### 13. Brick Vs Neighborhood Relationship:

To check the relationship of Brick and Non-Brick houses based on each neighborhood.

R Code : `Brick=as.factor(Brick)`

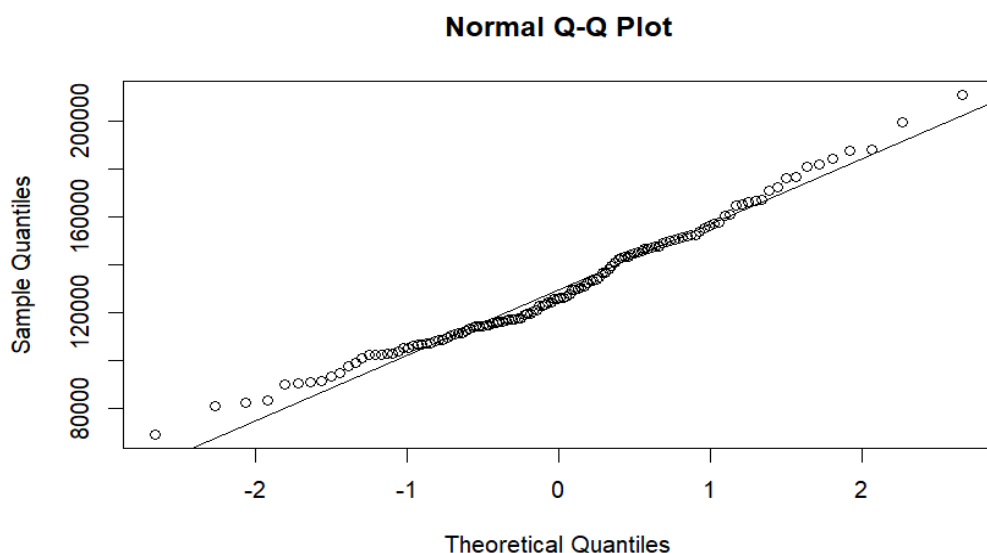
`plot(Brick~Neighborhood)`



**Analysis:** Based on the plot above we can check maximum no. of Non-Brick Houses (~80%) are located at north neighborhood. There are very few brick houses (~20%) on the North Neighborhood Region. Also, in the west and east region the houses are almost 40% Brick made and 60% non-brick made.

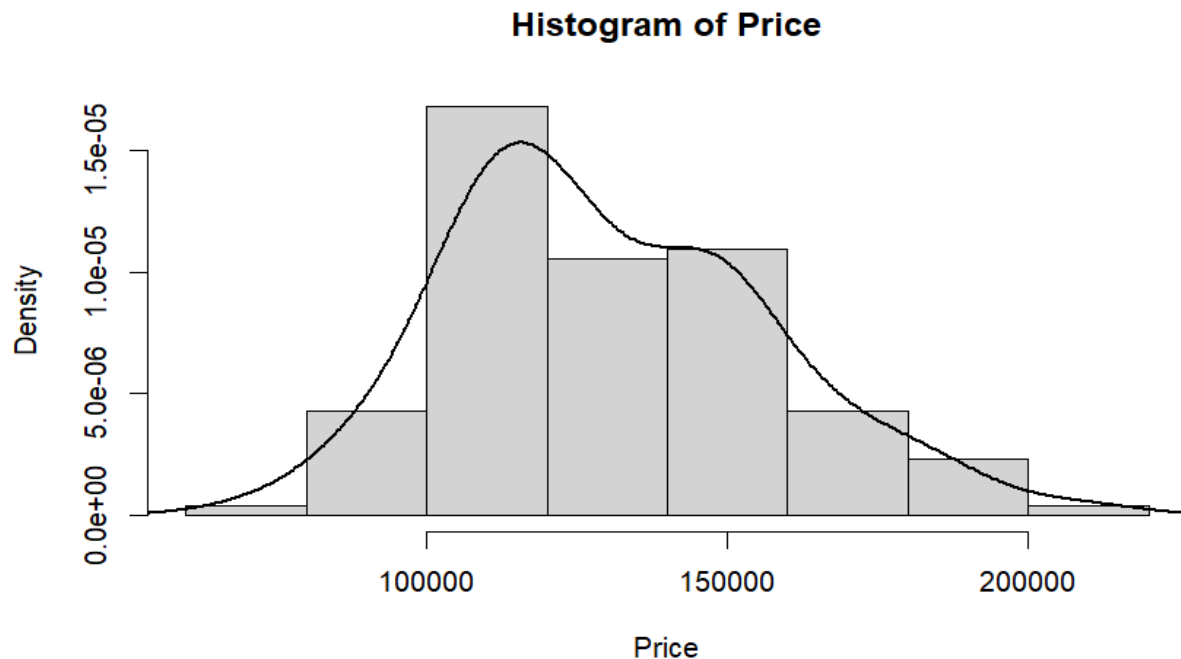
### 14. Normality check of Price, Neighborhood wise:

To check normality of Prices we have to check the qqnorm and qqline first.



From the above plot we can see most of the house prices are concentrated to the 1.S.D. of the plot.

Now we can plot the histogram of the data.



From the above plot we can conclude that the price data is not normally distributed as the curve is not a bell-shaped curve.

But to confirm the statement we can perform several statistical tests for the checking of the normality.

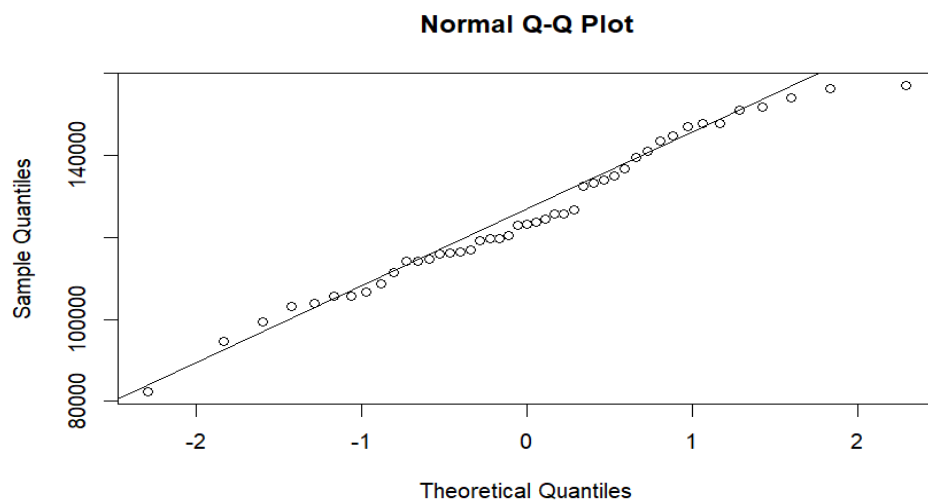
| Test Name | Shapiro-Wilk<br>normality test | Lilliefors<br>(Kolmogorov-<br>Smirnov) normality<br>test | Anderson-Darling<br>normality test |
|-----------|--------------------------------|----------------------------------------------------------|------------------------------------|
| P-value   | 0.05836                        | 0.01865                                                  | 0.02008                            |

From the above table, the Shapiro test suggest that the Price data have p value  $> 0.05$  but in Lillie Test and Ad Test the p value is  $< 0.05$ , which concludes that the price data is not normally distributed.

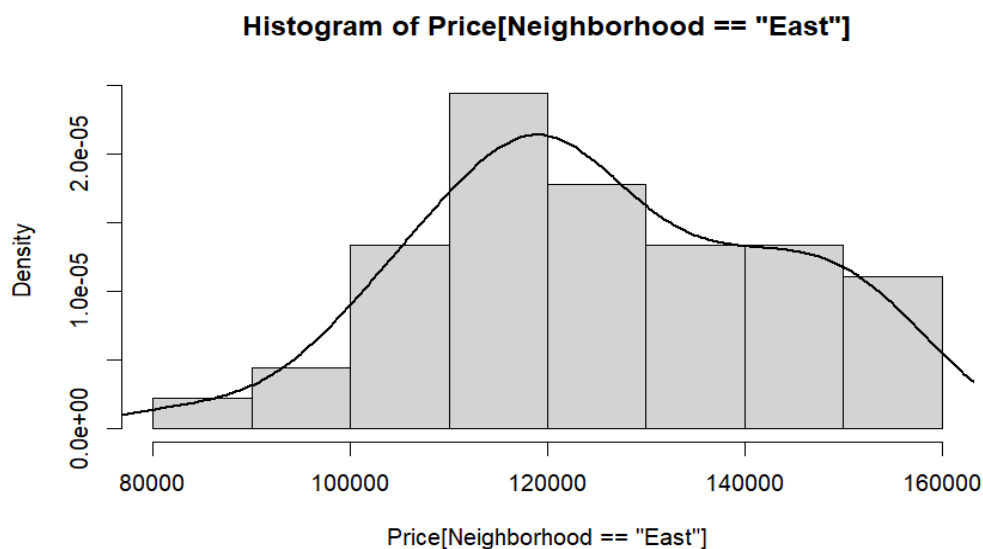
Now, we have to check normality of Price data, Neighborhood wise.

## Normality Check: House Prices: East neighborhood:

Checking qqline and qqnorm first-



Now we can plot the histogram-



From the above plot we can conclude that the price of the east neighborhood data is normally distributed as the curve is a bell-shaped curve but left skewed.

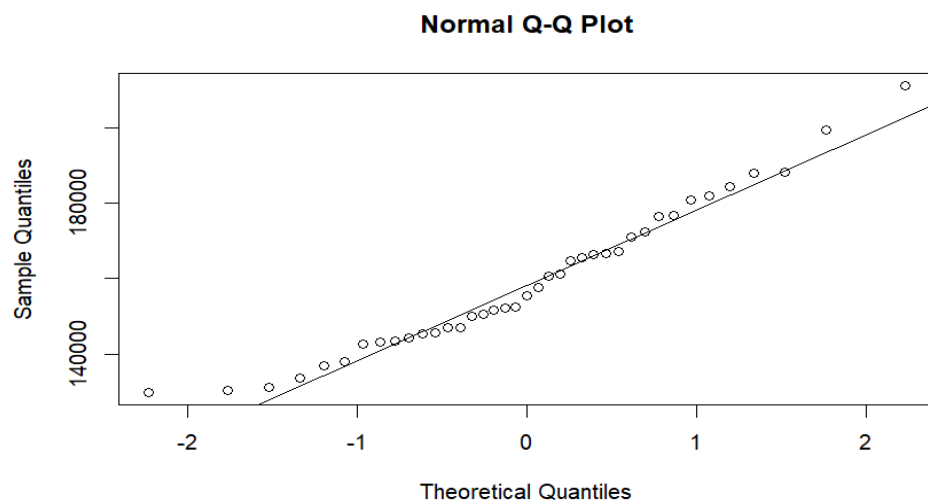
But to confirm the statement we can perform several statistical tests for the checking of the normality.

| Test Name | Shapiro-Wilk normality test | Lilliefors (Kolmogorov-Smirnov) normality test | Anderson-Darling normality test |
|-----------|-----------------------------|------------------------------------------------|---------------------------------|
| P-value   | 0.4661                      | 0.4872                                         | 0.4162                          |

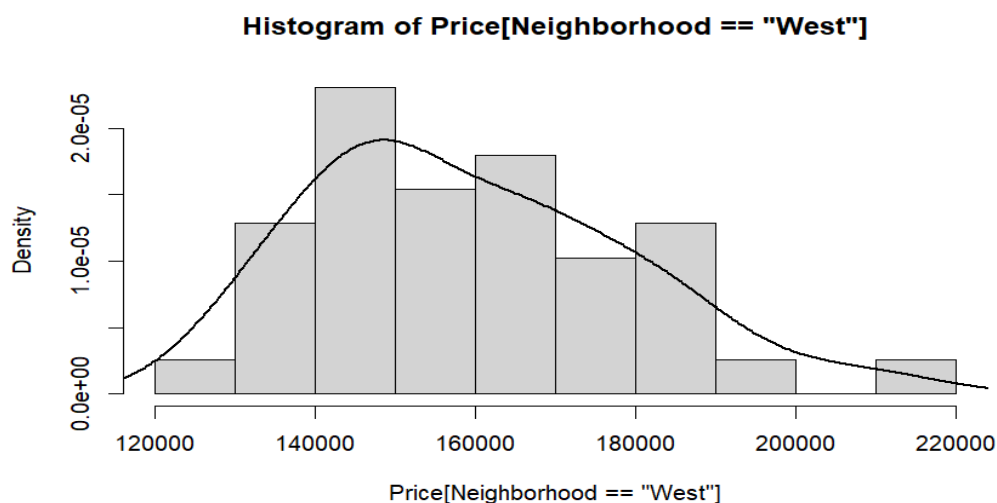
From the above table, all the test suggest that the Price data of East neighborhood have p value  $> 0.05$ , which concludes that the price data is normally distributed.

## Normality Check: House Prices: West neighborhood:

Checking qqline and qqnorm first-



Now we can plot the histogram-



From the above plot we can conclude that the price of the west neighborhood data is normally distributed as the curve is a bell-shaped curve but right skewed.

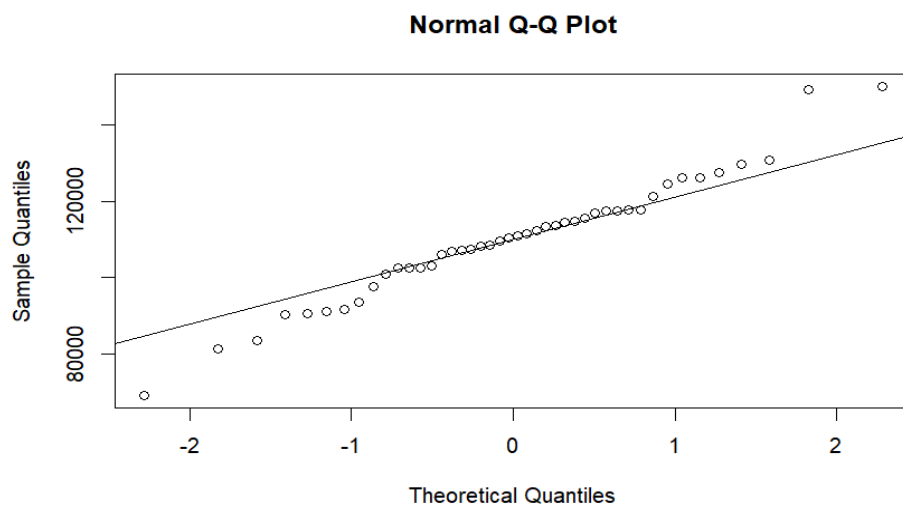
But to confirm the statement we can perform several statistical tests for the checking of the normality.

| Test Name | Shapiro-Wilk<br>normality test | Lilliefors<br>(Kolmogorov-<br>Smirnov) normality<br>test | Anderson-Darling<br>normality test |
|-----------|--------------------------------|----------------------------------------------------------|------------------------------------|
| P-value   | 0.1891                         | 0.1551                                                   | 0.2836                             |

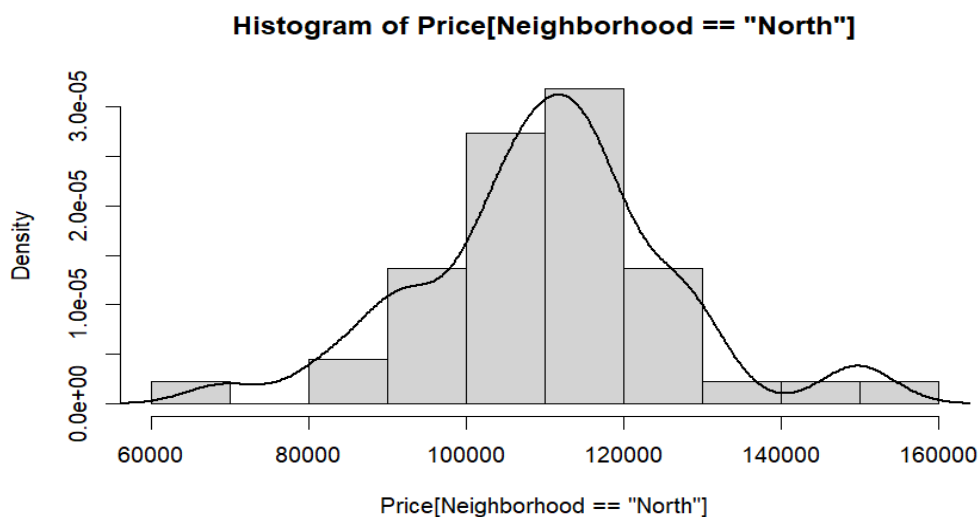
From the above table, all the test suggest that the Price data of west neighborhood have p value  $> 0.05$ , which concludes that the price data is normally distributed.

## Normality Check: House Prices: North neighborhood:

Checking qqline and qqnorm first-



Now we can plot the histogram-



From the above plot we can conclude that the price of the north neighborhood data is normally distributed as the curve is a bell-shaped curve but right skewed.

But to confirm the statement we can perform several statistical tests for the checking of the normality.

| Test Name | Shapiro-Wilk<br>normality test | Lilliefors<br>(Kolmogorov-<br>Smirnov) normality<br>test | Anderson-Darling<br>normality test |
|-----------|--------------------------------|----------------------------------------------------------|------------------------------------|
| P-value   | 0.4273                         | 0.1842                                                   | 0.2847                             |

From the above table, all the test suggest that the Price data of north neighborhood have p value  $> 0.05$ , which concludes that the price data is normally distributed.

### **15. Prediction of Price for future projects by the Promoter: (based on 1-sample t-test)**

Now from the previous data we can assume that the projects done by the promoter in the future will have an average price of 1,30,000. Let us check the assumption with help of 1-sample t-test.

$H_0$ : The avg. house prices of the projects will be 1,30,000.

$H_1$ : The avg. house prices of the projects will not be 1,30,000.

R code: `t.test(Price,mu=130000)`

#### **Output: One Sample t-test**

data: Price

$t = 0.17994$ ,  $df = 127$ ,  $p\text{-value} = 0.8575$

alternative hypothesis: true mean is not equal to 130000

95 percent confidence interval:

125727.9 135126.8

sample estimates:

mean of x

130427.3

**Analysis:** The p-value of the 1-sample t-test is 0.8575 which is much higher than 0.05 which implies that our null hypothesis is true and the future projects by the promoter will have an avg. house price of 1,30,000.

### **Prediction of future Price for the East Neighborhood projects by the Promoter: (based on 1-sample t-test):**

Now from the previous data we can assume that the future projects done by the promoter in the East Neighborhood will have an average price of 1,30,000. Let us check the assumption with help of 1-sample t-test.

$H_0$ : The avg. house prices of the East Neighborhood projects will be 1,30,000.

$H_1$ : The avg. house prices of the East Neighborhood projects will not be 1,30,000.

R code: `t.test(Price[Neighborhood=="East"],mu=130000)`

#### **Output: One Sample t-test**

data: Price[Neighborhood == "East"]

$t = -1.7906$ ,  $df = 44$ ,  $p\text{-value} = 0.08024$

alternative hypothesis: true mean is not equal to 130000

95 percent confidence interval: 119863.6 130598.7

sample estimates: mean of x: 125231.1

**Analysis:** The p-value of the 1-sample t-test is 0.08024 which is higher than 0.05 which implies that our null hypothesis is true and the future projects in the East Neighbourhood by the promoter will have an avg. house price around 1,30,000.

**Prediction of future Price for the West Neighborhood projects by the Promoter: (based on 1-sample t-test):**

Now from the previous data we can assume that the future projects done by the promoter in the West Neighborhood will have an average price of 1,30,000. Let us check the assumption with help of 1-sample t-test.

$H_0$ : The avg. house prices of the West Neighborhood projects will be 1,30,000.

$H_1$ : The avg. house prices of the West Neighborhood projects will not be 1,30,000.

R code: `t.test(Price[Neighborhood=="West"],mu=130000)`

**Output: One Sample t-test:**

data: Price[Neighborhood == "West"]

t = 9.2483, df = 38, p-value = 2.861e-11

alternative hypothesis: true mean is not equal to 130000

95 percent confidence interval: 152882.4 165707.4

sample estimates: mean of x: 159294.9

**Analysis:** The p-value of the 1-sample t-test is 2.861e-11 which is lesser than 0.05 which implies that our null hypothesis is false and the alternate hypothesis is true. Future projects in the West Neighbourhood by the promoter will not have an avg. house price around 1,30,000. The avg. house price of the west neighborhood will have an avg. in the range of 152882.4-165707.4.

**Prediction of future Price for the North Neighborhood projects by the Promoter: (based on 1-sample t-test):**

Now from the previous data we can assume that the future projects done by the promoter in the North Neighborhood will have an average price of 1,30,000. Let us check the assumption with help of 1-sample t-test.

$H_0$ : The avg. house prices of the North Neighborhood projects will be 1,30,000.

$H_1$ : The avg. house prices of the North Neighborhood projects will not be 1,30,000.

R code: `t.test(Price[Neighborhood=="North"],mu=130000)`

**Output: One Sample t-test:**

data: Price[Neighborhood == "North"]

t = -8.2409, df = 43, p-value = 2.166e-10

alternative hypothesis: true mean is not equal to 130000

95 percent confidence interval: 105298 115011

sample estimates: mean of x: 110154.5

**Analysis:** The p-value of the 1-sample t-test is  $2.166e-10$  which is lesser than 0.05 which implies that our null hypothesis is false and the alternate hypothesis is true. Future projects in the West Neighbourhood by the promoter will not have an avg. house price around 1,30,000. The avg. house price of the west neighborhood will have an avg. in the range of 105298-115011.

#### **16. Variance in the Price based on Bricks: (based on 2-sample t-test) :**

There is difference between the prices between brick-based houses and non-brick-based houses.

R code: `Brick=as.factor(Brick)`

`levene.test(Price~Brick)`

`t.test(Price~Brick,var.equal = T)`

#### **Two Sample t-test**

Data: Price by Brick

$t = -5.7008$ ,  $df = 126$ ,  $p\text{-value} = 8.023e-08$

Alternative hypothesis: True difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

-34770.88                      -16850.94

Sample estimates:

| Mean in group Brick=No | Mean in group Brick=Yes |
|------------------------|-------------------------|
| 121958.1               | 147769.0                |

**Analysis:** There is significant difference between the avg. prices between Brick based houses and non-brick-based houses. The maximum difference between prices of two types of houses is 34770 and minimum difference is 16850.



## 17. Price per SqFt relationship analysis:

We can analyze the price per square ft. relationship on the basis of the following data.

We can find the mean, median value of Price per sq.ft., plot the histogram, qqnorm, qqline, density line, and analysis on the basis of that.

R code:  $\text{PriceperSqFt} = \text{Price} / \text{SqFt}$

```
summary(PriceperSqFt)
```

```
boxplot(PriceperSqFt~Neighborhood, horizontal = T)
```

```
hist(PriceperSqFt, freq = F)
```

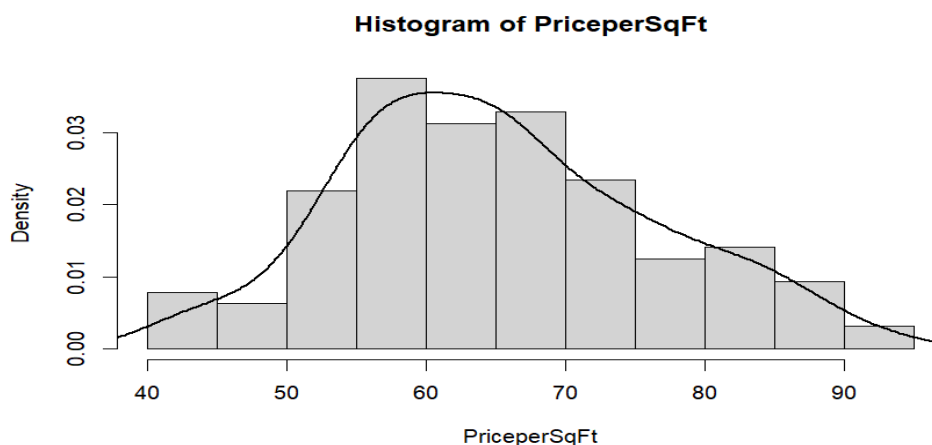
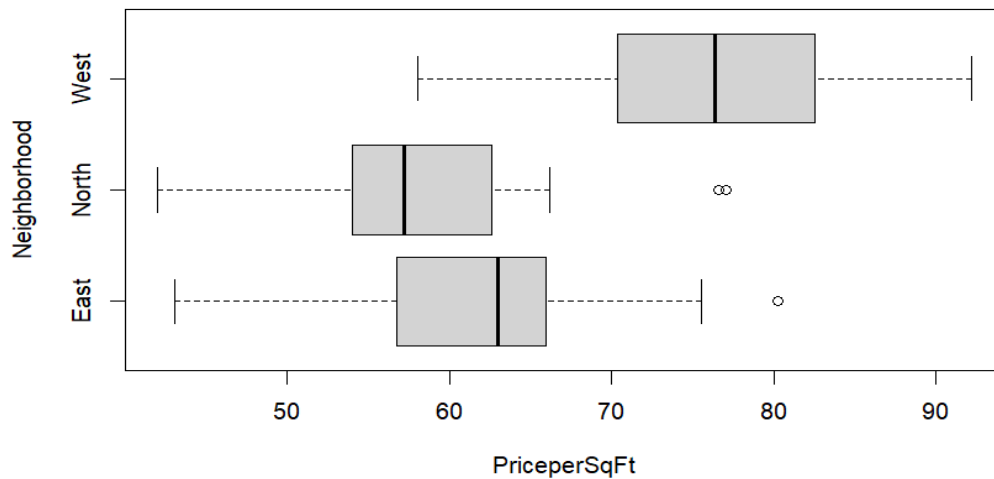
```
lines(density(PriceperSqFt), lwd=2)
```

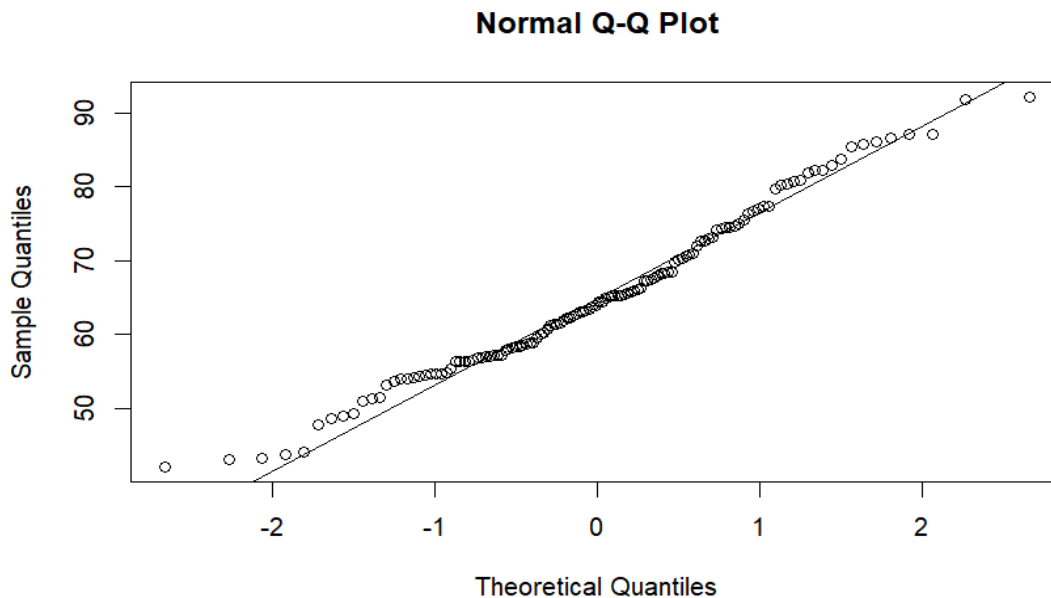
```
qqnorm(PriceperSqFt)
```

```
qqline(PriceperSqFt)
```

Output: `summary(PriceperSqFt)`

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 42.01 | 57.00   | 64.10  | 65.13 | 72.73   | 92.16 |





Now we can perform the statistical tests to check the normality of the data of Price per SqFt.

| Test Name | Shapiro-Wilk<br>normality test | Lilliefors<br>(Kolmogorov-<br>Smirnov) normality<br>test | Anderson-Darling<br>normality test |
|-----------|--------------------------------|----------------------------------------------------------|------------------------------------|
| P-value   | 0.06462                        | 0.09954                                                  | 0.03852                            |

R code: `shapiro.test(PriceperSqFt)`

`ad.test(PriceperSqFt)`

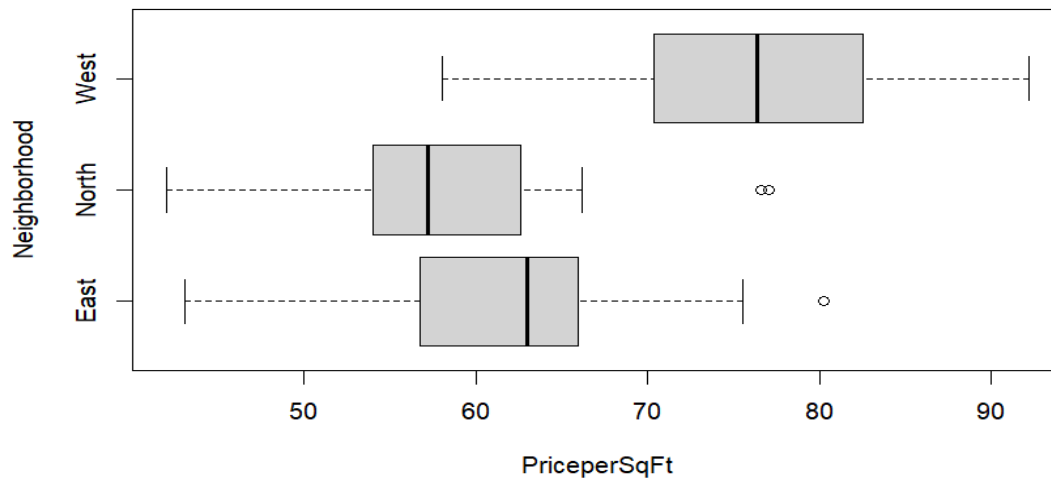
`lillie.test(PriceperSqFt)`

**Analysis:** From the above data we can check from Shapiro test and Lillie test, p-value > 0.05 and in Lillie test the p-value is < 0.05, so we can conclude that the per square ft value have normal distribution and the density line is normally distributed. The avg. per sq. ft price is 65.13 and median 64.10. Lastly the west neighborhood has more per sq. ft prices than other two neighborhoods.

**18. Neighborhood Vs Price per SqFt Relationship:** We can check the price per sq.ft with respect to the neighborhood using the boxplot.

R code: `Neighborhood=as.factor(Neighborhood)`

`boxplot(PriceperSqFt~Neighborhood, horizontal = T)`



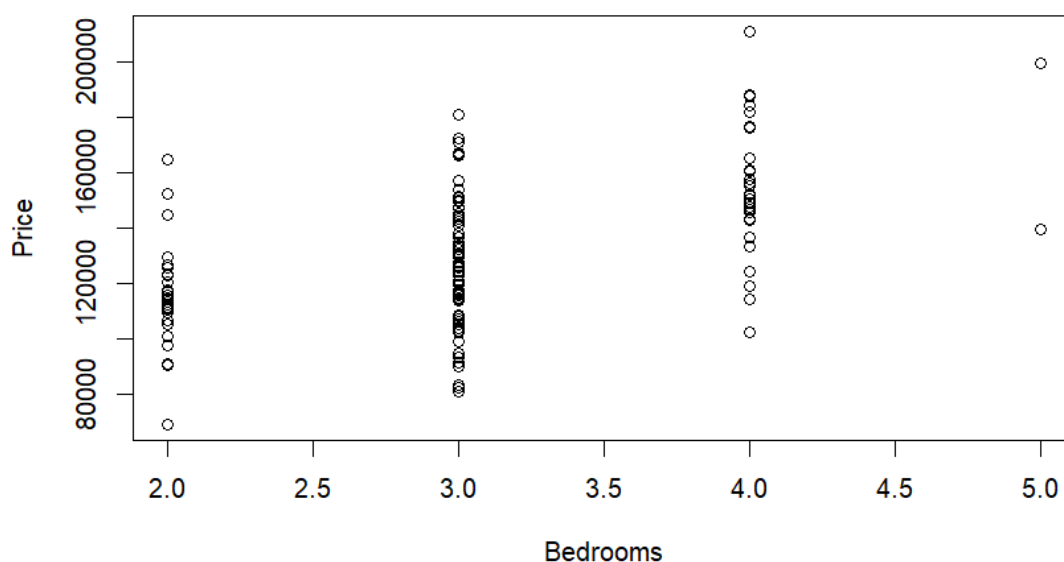
**Analysis:** Price per SqFt is much higher in the west neighborhood as compare to the other neighborhoods. West neighborhood area is more prior location so have much higher price per Sqft valuation.

### **19. Relationship Between Price Vs Bedrooms:**

The no. of bedrooms and house prices have a positive co-relationship value of 0.5259. This implies that they have a positive co relationship between them and as no. of bedrooms increases the price of the houses also increases.

R code: `plot(Price~Bedrooms)`

`cor(Price,Bedrooms)`



**Analysis:** Maximum no. of houses have 2 or 3 bedrooms.

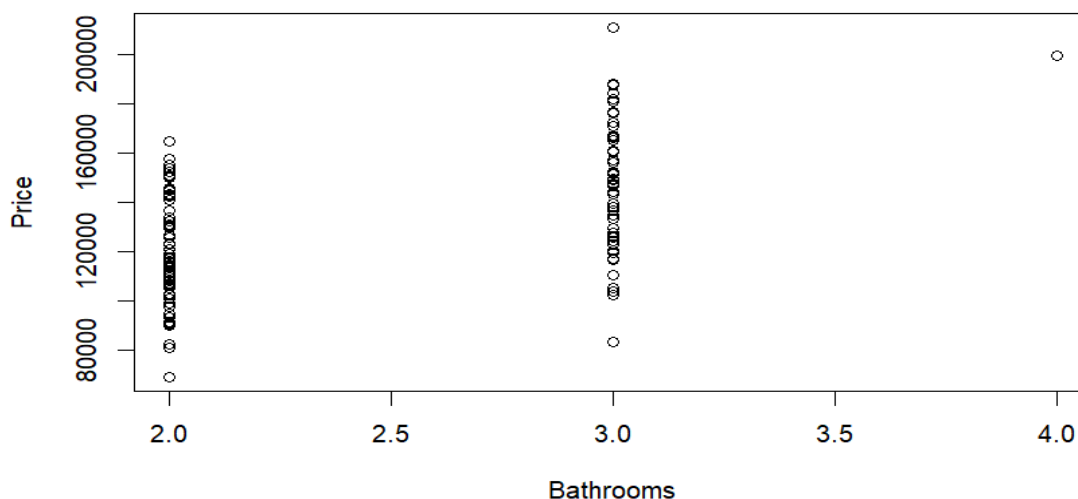
| No. of bedrooms | 2 bedrooms | 3 bedrooms | 4 bedrooms |
|-----------------|------------|------------|------------|
| Mean Price      | 115260     | 125732.8   | 154265.5   |

## **20. Relationship between Price Vs Bathrooms:**

The no. of bathrooms and house prices have a positive co-relationship value of 0.5233. This implies that they have a positive co relationship between them and as no. of bathrooms increases the price of the houses also increases.

R code: `plot(Price~Bathrooms)`

`cor(Price,Bathrooms)`



**Analysis:** From the plot, we can say that maximum no. of houses either have 2 or 3 bathrooms.

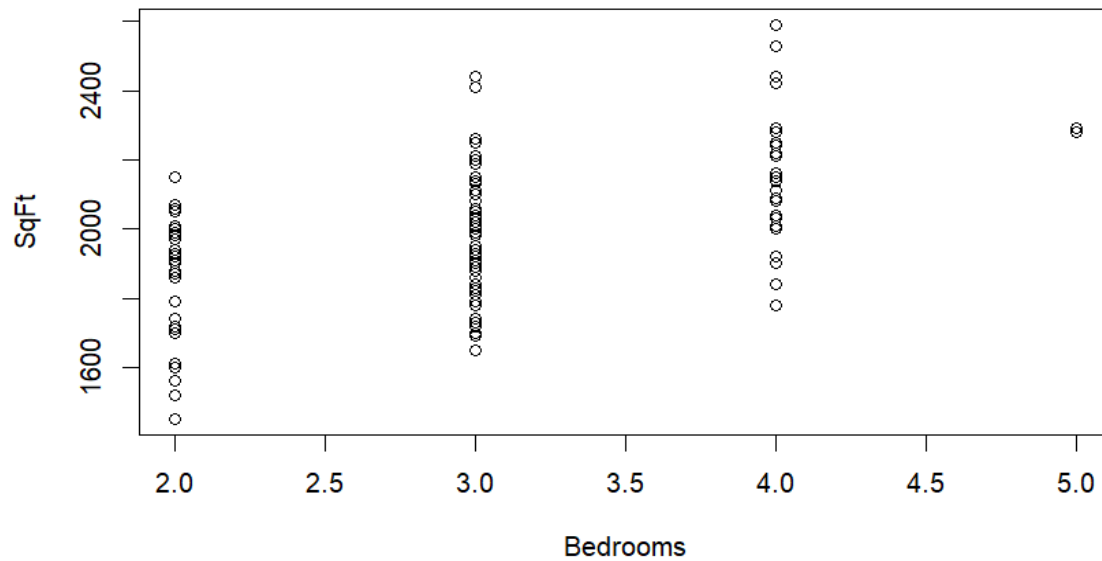
| No. of bathrooms | 2 bathrooms | 3 bathrooms |
|------------------|-------------|-------------|
| Mean Price       | 118627.8    | 144618.2    |

## **21. SqFt vs Bedroom Relationship:**

The no. of bedrooms and SqFt have a positive co-relationship value of 0.4838. This implies that they have a positive co relationship between them and as no. of bedrooms increases, the SqFt area of the houses also increases.

R code: `plot(SqFt~Bedrooms)`

`cor(SqFt,Bedrooms)`



**Analysis:** From the plot it is clear that maximum houses have 3 bedrooms within the SqFt range of 1800-2200 sq.ft.

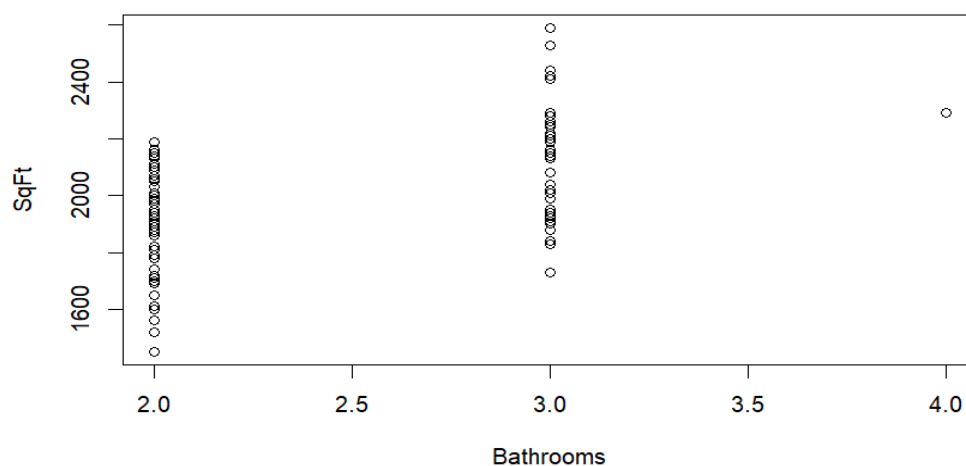
| No. of bedrooms | 2 bedrooms | 3 bedrooms | 4 bedrooms |
|-----------------|------------|------------|------------|
| Mean SqFt area  | 1860.33    | 1994.478   | 2141.724   |

## **22. SqFt Vs Bathrooms Relationship:**

The no. of bathrooms and SqFt have a positive co-relationship value of 0.5227. This implies that they have a positive co relationship between them and as no. of bathrooms increases, the SqFt area of the houses also increases.

R code: `plot(SqFt~Bathrooms)`

`cor(SqFt,Bathrooms)`

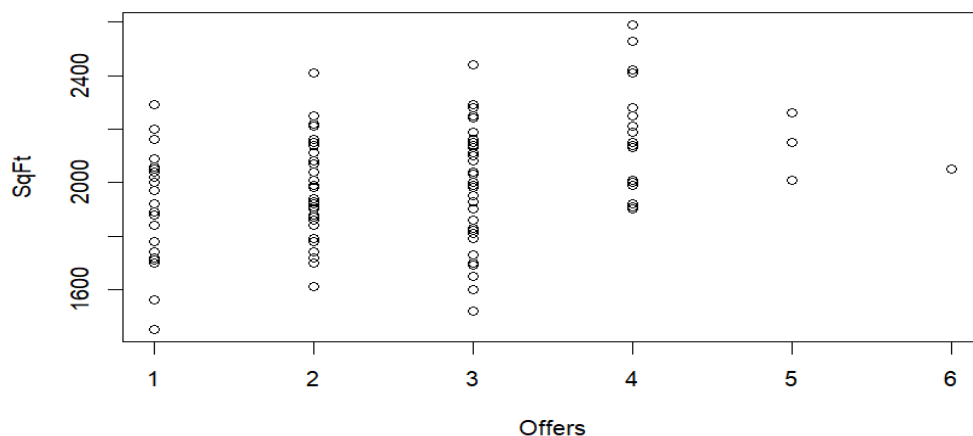


**Analysis:** From the plot it is clear that maximum houses have 2 or 3 bathrooms within the SqFt range of 1800-2200 sq.ft.

|                         |                    |                    |
|-------------------------|--------------------|--------------------|
| <b>No. of bathrooms</b> | <b>2 bathrooms</b> | <b>3 bathrooms</b> |
| <b>Mean SqFt</b>        | <b>1904.583</b>    | <b>2121.818</b>    |

### **23. Relationship between SqFt and Offers:**

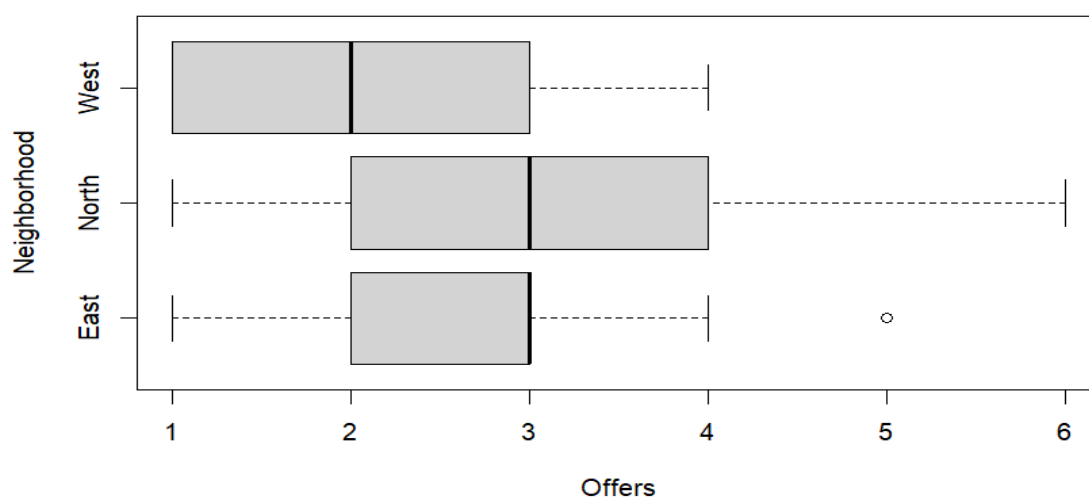
The co-relationship between SqFt and offers is 0.3369 implies that they have a positive co-relationship and SqFt area increases, no. of offers also increases.



### **24. Neighborhood Vs Offer Relationship:**

From the plot, we can say the west neighborhood has less no. of offers whereas the north neighborhood has the maximum no. of offers.

R code: `Neighborhood=as.factor(Neighborhood)`  
`plot(Offers~Neighborhood,horizontal=T)`



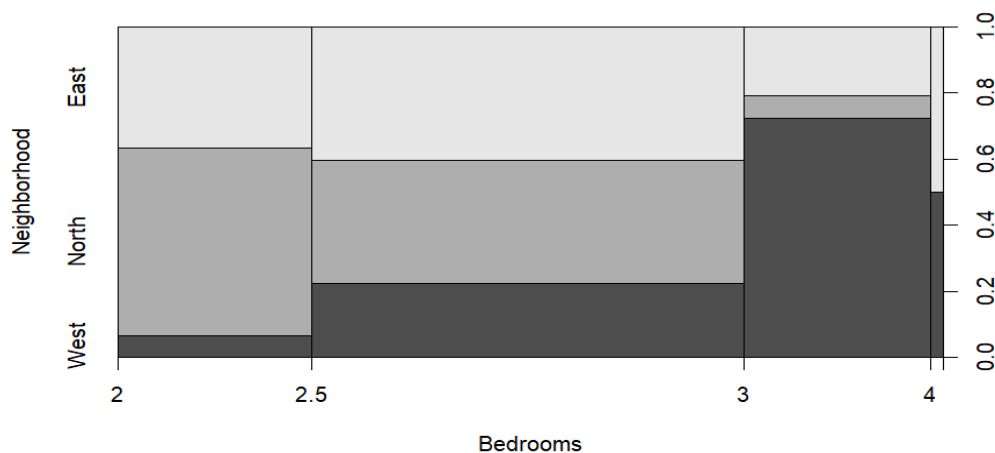
**Analysis:** analyzing the above plot we can conclude that as the price of the north neighborhood is less and the demand of the north neighborhood houses are less that's why the promoter has to provide more offers in that region and sell the north neighborhood houses at a lower price.

## **25. Neighborhood Vs Bedroom Relationship:**

From the plot, we can say the west neighborhood has more bedroom houses with larger areas whereas the north neighborhood has the less no of bedroom houses with lesser area.

R code: `Neighborhood=as.factor(Neighborhood)`

`plot(Neighborhood~Bedrooms,horizontal=T)`

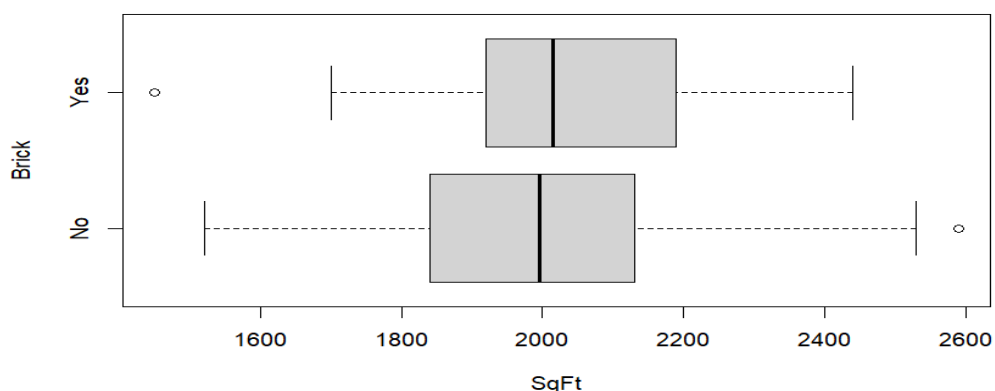


**Analysis:** The west neighborhood has houses with more bedrooms and the north neighborhood has the houses with lesser no. of bedrooms. That's why the west neighborhood houses are higher in price than the north neighborhood houses.

## **26. SqFt Vs Brick Relationship:**

From the plot we can check that the median range of SqFt for brick and non-brick houses are almost same but the SqFt range of the non-brick houses are much more than the Brick made houses.

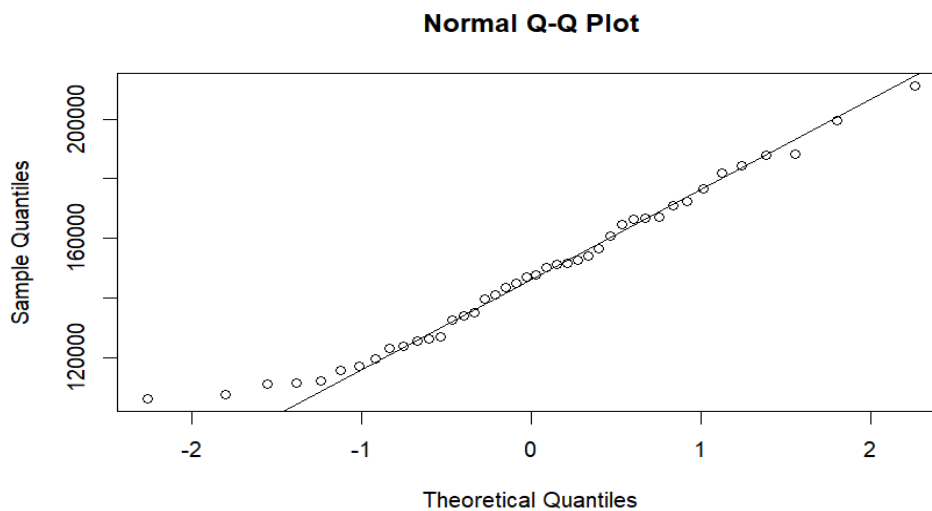
R code: `plot(SqFt~Brick,horizontal=T)`



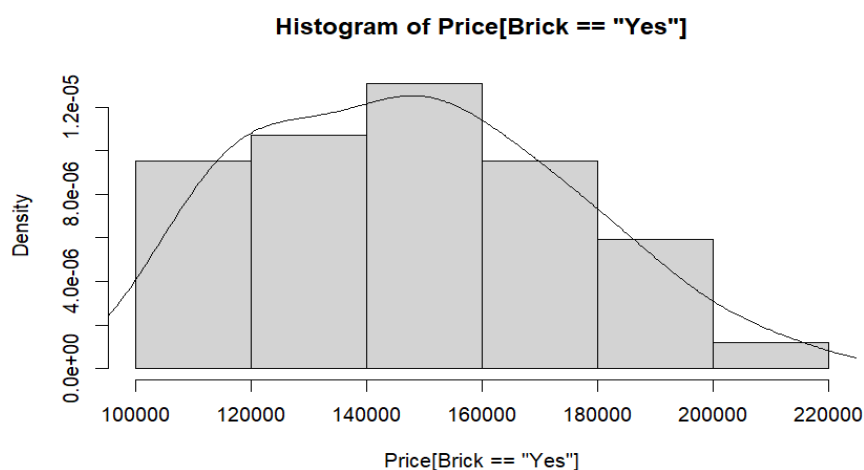
## 27: Normality check of House Price based on Bricks:

### Normality Check: House Prices: Brick=="Yes":

Checking qqline and qqnorm first-



Now we can plot the histogram-



From the above plot we can conclude that the price of the Brick based houses are normally distributed as the curve is a bell-shaped curve but right skewed.

But to confirm the statement we can perform several statistical tests for the checking of the normality.

| Test Name | Shapiro-Wilk<br>normality test | Lilliefors<br>(Kolmogorov-<br>Smirnov) normality<br>test | Anderson-Darling<br>normality test |
|-----------|--------------------------------|----------------------------------------------------------|------------------------------------|
| P-value   | 0.3379                         | 0.4919                                                   | 0.5674                             |

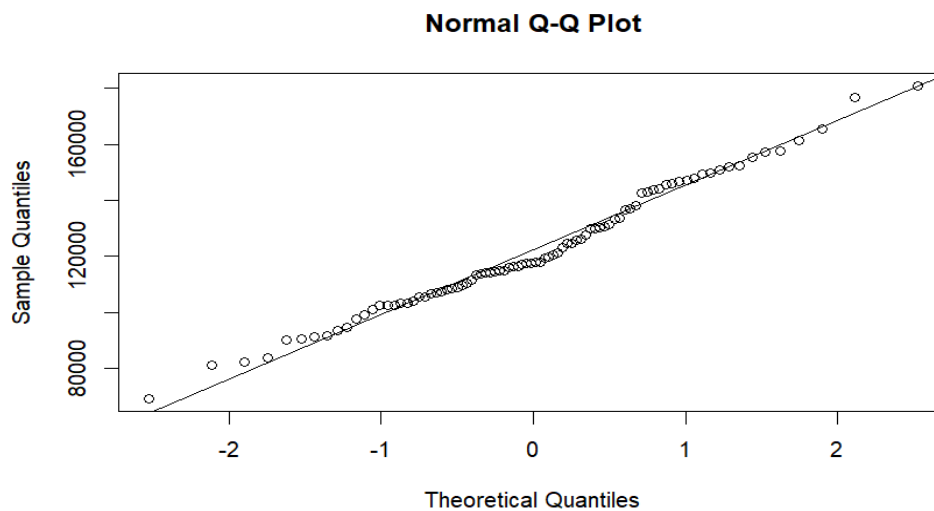
From the above table, all the test suggest that the Price data of Brick based houses have p value  $> 0.05$ , which concludes that the brick-based price data is normally distributed.



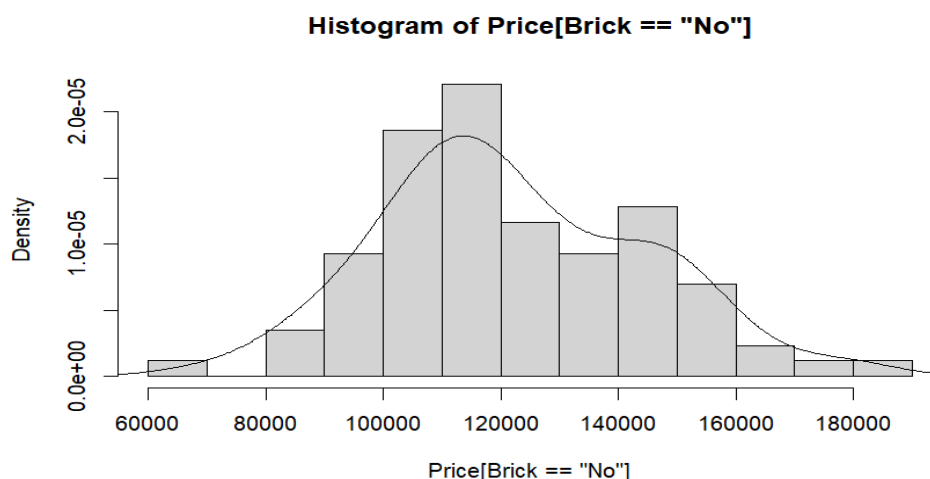
## 28: Normality check of House Price based on Bricks:

### Normality Check: House Prices: Brick=="No":

Checking qqline and qqnorm first-



Now we can plot the histogram-



From the above plot we can conclude that the price of the non-brick based houses are normally distributed as the curve is a bell-shaped curve but right skewed.

But to confirm the statement we can perform several statistical tests for the checking of the normality.

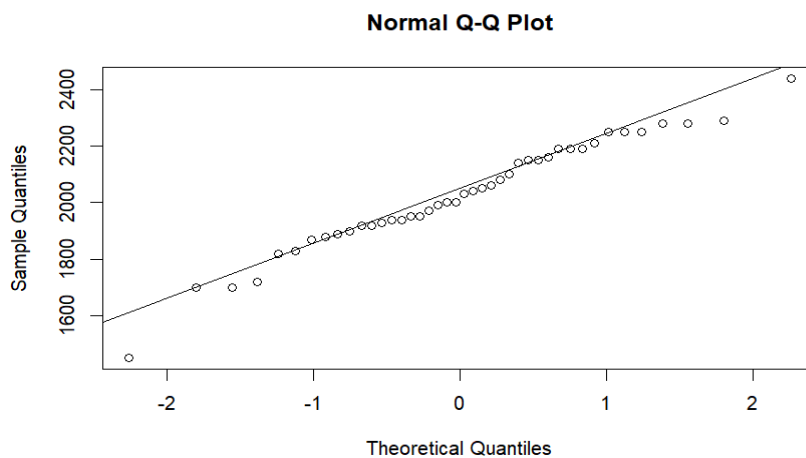
| Test Name | Shapiro-Wilk normality test | Lilliefors (Kolmogorov-Smirnov) normality test | Anderson-Darling normality test |
|-----------|-----------------------------|------------------------------------------------|---------------------------------|
| P-value   | 0.4399                      | 0.04732                                        | 0.1545                          |

Here, Shapiro test and Ad-test suggest that the Price data of non-brick based houses have p value  $> 0.05$ , which concludes that the non-brick based price data is normally distributed.

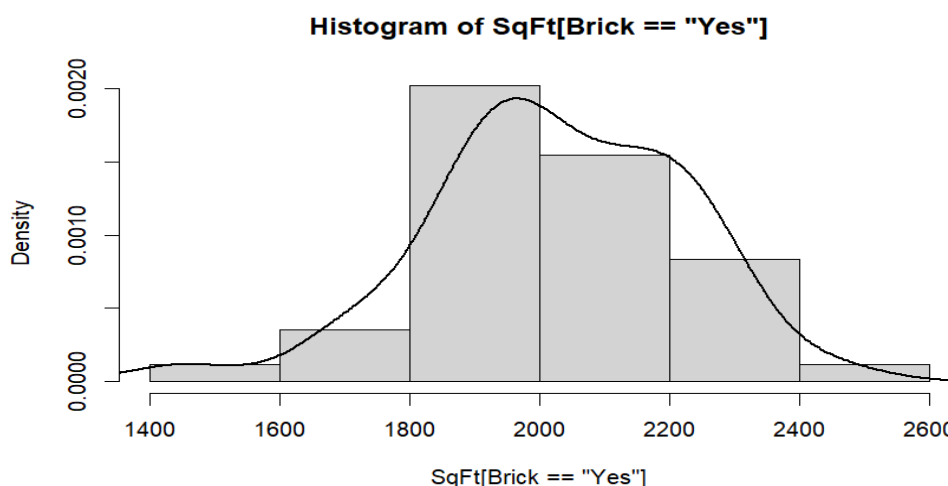
## 29. Normality check of SqFt based on Bricks:

### Normality Check: SqFt: Brick=="Yes":

Checking qqline and qqnorm first-



Now we can plot the histogram-



From the above plot we can conclude that the SqFt data of the Brick based houses are normally distributed as the curve is a bell-shaped curve but left skewed.

But to confirm the statement we can perform several statistical tests for the checking of the normality.

```
R code: qqnorm(SqFt[Brick=="Yes"])
        qqline(SqFt[Brick=="Yes"])
        hist(SqFt[Brick=="Yes"],freq = F)
        lines(density(SqFt[Brick=="Yes"]),lwd=2)
        shapiro.test(SqFt[Brick=="Yes"])
        lillie.test(SqFt[Brick=="Yes"])
        ad.test(SqFt[Brick=="Yes"])
```

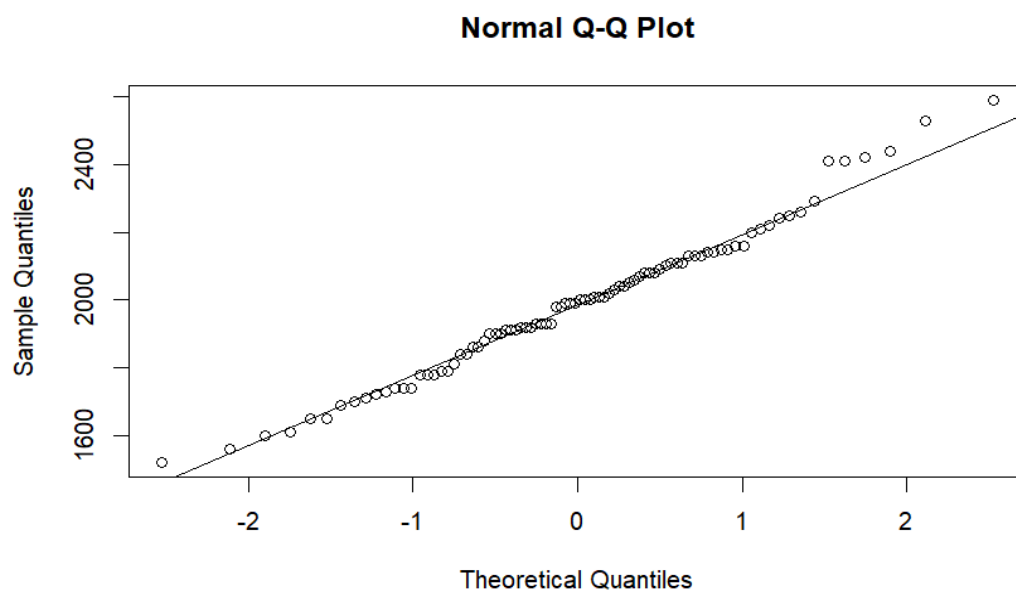
| Test Name | Shapiro-Wilk normality test | Lilliefors (Kolmogorov-Smirnov) normality test | Anderson-Darling normality test |
|-----------|-----------------------------|------------------------------------------------|---------------------------------|
| P-value   | 0.4598                      | 0.7375                                         | 0.5142                          |

Here, all the statistical tests suggest that the SqFt data of brick-based houses have p value > 0.05, which concludes that the brick based SqFt data is normally distributed.

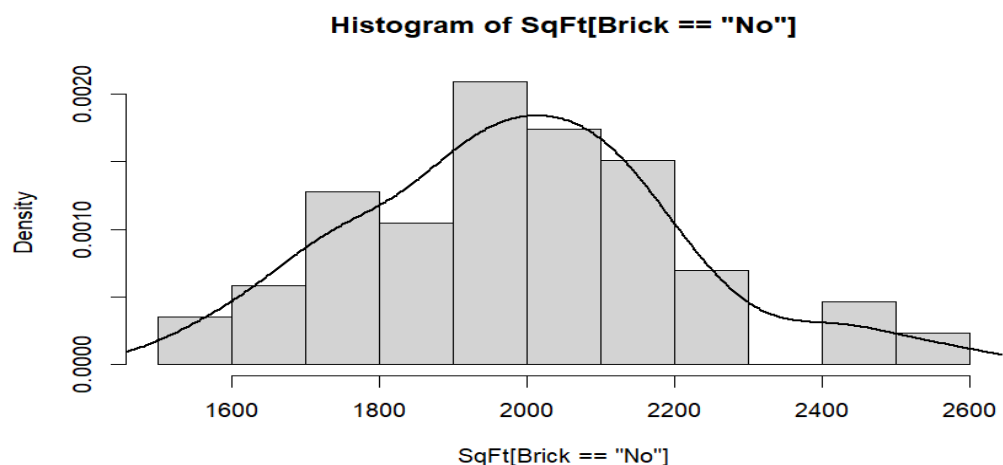
### 30. Normality check of SqFt based on Bricks:

**Normality Check: SqFt: Brick=="No":**

Checking qqline and qqnorm first-



Now we can plot the histogram-



From the above plot we can conclude that the SqFt data of the Brick based houses are normally distributed as the curve is a bell-shaped curve but right skewed.

But to confirm the statement we can perform several statistical tests for the checking of the normality.

```
R code: qqnorm(SqFt[Brick=="No"])
        qqline(SqFt[Brick=="No"])
        hist(SqFt[Brick=="No"],freq = F)
        lines(density(SqFt[Brick=="No"]),lwd=2)
        shapiro.test(SqFt[Brick=="No"])
        lillie.test(SqFt[Brick=="No"])
        ad.test(SqFt[Brick=="No"])
```

| Test Name | Shapiro-Wilk<br>normality test | Lilliefors<br>(Kolmogorov-<br>Smirnov) normality<br>test | Anderson-Darling<br>normality test |
|-----------|--------------------------------|----------------------------------------------------------|------------------------------------|
| P-value   | 0.4661                         | 0.4553                                                   | 0.5202                             |

Here, all the statistical tests suggest that the SqFt data of brick-based houses have p value > 0.05, which concludes that the brick based SqFt data is normally distributed.

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---The End---